Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 1

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. Students should take the signature from the faculty in charge. For every laboratory session, the observation component carrie s**marks** .

Problem Statements :

1. The Zombie Bridge Crossing (Greedy / Min-Max Optimization)

- **Problem Statement:** Taking an internship at a secluded mountain lab seemed like a great idea until you accidentally pull a lever that unleashes a horde of zombies. You, an Assistant, a Janitor, and an ancient Professor must escape across a rickety bridge at night.
    - The bridge can hold at most 2 people at a time.
    - You have only 1 lantern, which is required for anyone crossing.
    - Crossing times: You (1 minute), Assistant (2 minutes), Janitor (5 minutes), Professor (10 minutes).
    - When two people cross together, they move at the speed of the slower person.
- **Question:** What is the minimum total time required for all 4 people to safely cross the bridge before the zombies reach it?
- **Answer & Explanation:**
    - Step 1. You and the Assistant cross together with the lantern (Time elapsed: 2 minutes).
    - Step 2. You run back across the bridge holding the lantern (Time elapsed: 2 + 1 = 3 minutes).
    - Step 3. The two slowest people (Janitor and Professor) cross together with the lantern (Time elapsed: 3 + 10 = 13 minutes).
    - Step 4. The Assistant takes the lantern and runs back across to meet you (Time elapsed: 13 + 2 = 15 minutes).
    - Step 5. You and the Assistant cross together a final time (Time elapsed: 15 + 2 = 17 minutes). All 4 people are safe in 17 minutes.

- **Animated Video:** Can you solve the bridge riddle? - Alex Gendler

2. The Counterfeit Gold Coin (Ternary Division / Decision Trees)

- **Problem Statement:** An emperor's treasury contains 12 gold coins, but 1 of the coins is counterfeit and weighs slightly less than the genuine coins. You are given a balance scale that compares the weight of two sets of coins.

- **Question:** What is the minimum number of balance scale weighings needed to guarantee finding the counterfeit coin?

- **Answer & Explanation:**

    - Step 1. Divide the 12 coins into 3 equal groups of 4 coins each (Group A, Group B, Group C).

    - Step 2. Weigh Group A against Group B (Weighing 1). If they balance, the counterfeit coin is in Group C. If one side is lighter, the counterfeit is in that group.

    - Step 3. Take the 4 suspected coins and place 1 coin on each side of the scale (Weighing 2).

    - Step 4. If they balance, weigh the remaining 2 unweighed coins against each other (Weighing 3) to isolate the lighter fake coin.

    - Step 5. Hence, dividing by 3 at each stage guarantees locating the fake coin in 3 weighings.

- **Animated Video:** Can you solve the counterfeit coin riddle? - Jennifer Lu

3. The 100 Uncle's Lockers (Number Theory / Perfect Squares)

- **Problem Statement:** An eccentric uncle leaves a row of 100 closed lockers to 100 heirs.

    - Person 1 walks down the row and opens every locker.

    - Person 2 walks down and toggles (closes) every 2nd locker.

    - Person 3 toggles every 3rd locker, and so on, until Person 100 toggles the 100th locker.

- **Question:** Which lockers remain open after all 100 people have finished toggling?

- **Answer & Explanation:**
    - Step 1. A locker is toggled once for every factor its locker number possesses.
    - Step 2. Most positive integers have an even number of factors (paired as $a \times b = N$), causing the locker to end up in its initial closed state.
    - Step 3. Only perfect square numbers $(1,4,9,16,25,36,49,64,81,100)$ have an odd number of factors because their square root factor pairs with itself.
    - Step 4. Because they are toggled an odd number of times, exactly 10 lockers (all perfect squares from 1 to 100) remain open at the end.
- **Animated Video:** Can you solve the locker riddle? - Lisa Winer

4. The Band Member Instrument Boxes (Cycle Search / Permutation Cycles)

- **Problem Statement:** 10 band members have their musical instruments randomly placed inside 10 closed boxes. Each musician enters the room individually and can open at most 5 boxes to find their assigned instrument before closing all boxes. They cannot communicate once the process begins. If all 10 members find their own instruments, they succeed; if even one fails, everyone loses.
- **Question:** What strategy gives the band a success rate of over 30% instead of random guessing (0.1%)?
- **Answer & Explanation:**
    - Step 1. Each musician goes to the box labeled with their own assigned number first.
    - Step 2. The musician opens that box, looks at the instrument inside, and goes to the box corresponding to that instrument's label.
    - Step 3. By following this pointer chain, the musician follows a closed permutation cycle.
    - Step 4. All musicians succeed as long as the random arrangement contains no permutation cycles longer than 5 boxes.
    - Step 5. The probability that a random permutation of 10 items has no cycle longer than 5 is approximately 35.4%.

- **Animated Video:** Can you solve the prisoner boxes riddle? - Yossi Elran

5. The Pirate Booty Division (Game Theory / Backward Induction)

- **Problem Statement:** 5 pirates (ranked in seniority 1 to 5) must divide 100 gold coins. The senior-most pirate proposes a distribution plan. All living pirates vote. If at least 50% approve, the plan is executed; otherwise, the proposer is thrown overboard, and the next senior pirate proposes. All pirates are rational, greedy, and bloodthirsty.

- **Question:** How should Pirate 5 (the most senior) propose the split to maximize gold while surviving?

- **Answer & Explanation:**

  - Step 1. Solve backward: If only Pirates 1 and 2 remain, Pirate 2 takes 100 coins and votes for himself (50% vote pass). Pirate 1 gets 0.

  - Step 2. If 3 remain: Pirate 3 offers Pirate 1 one coin ($1 > 0$) to secure Pirate 1's vote. Split: [1: 1, 2: 0, 3: 99].

  - Step 3. If 4 remain: Pirate 4 offers Pirate 2 one coin ($1 > 0$) to secure Pirate 2's vote. Split: [1: 0, 2: 1, 3: 0, 4: 99].

  - Step 4. If 5 remain: Pirate 5 needs 2 votes besides his own. He buys Pirate 1 (1 coin) and Pirate 3 (1 coin).

  - Step 5. Optimal allocation for Pirate 5: 98 coins for himself, 1 coin for Pirate 3, and 1 coin for Pirate 1 ([98, 0, 1, 0, 1]).

- **Animated Video:** Can you solve the pirate riddle? - Alex Gendler

6. The Airborne Virus Containment (Graph Traversal / Hamiltonian Path)

- **Problem Statement:** An earthquake breaks virus vials in a 4×4 grid of 16 laboratory rooms. You enter at the top-left corner room (Room 1) and must pass through every contaminated room to destroy the virus before escaping through the exit at the bottom-left corner room (Room 13). You cannot re-enter a room once purified.

- **Question:** What path allows you to visit all 16 rooms exactly once and end at the exit room?

- **Answer & Explanation:**
    - Step 1. Model the grid as a bipartite graph where adjacent rooms alternate colors like a chessboard.
    - Step 2. The start room (1) and exit room (13) share the same color tile.
    - Step 3. A path visiting all 16 rooms must alternate colors 15 times, meaning the start and end rooms MUST be of opposite colors.
    - Step 4. Because Room 1 and Room 13 are the same color, a standard Hamiltonian path of length 16 is mathematically impossible unless a room's state is modified.
    - Step 5. Destroying the virus in Room 1 immediately upon entry allows you to step back into Room 1's airlock, modifying the path structure to solve the traversal cleanly.

- **Animated Video:** Can you solve the virus riddle? - Lisa Winer

7. The Fabergé Egg Drop (Dynamic Programming / Min-Max Drops)

- **Problem Statement:** A museum thief has 2 identical Fabergé eggs and wants to find the highest floor of a 100-story skyscraper from which an egg can be dropped without breaking. If an egg breaks, it cannot be dropped again.

- **Question:** What is the minimum number of drops required in the worst-case scenario to find the exact threshold floor?

- **Answer & Explanation:**
    - Step 1. Dropping from constant increments (e.g., every 10th floor) results in up to 19 drops worst-case ($10 + 9$).
    - Step 2. To keep total drops constant across all outcomes, reduce the stride length by 1 for each subsequent drop: $x + (x-1) + (x-2) + \cdots + 1 \geq 100$.
    - Step 3. Solve equation: $\frac{x(x+1)}{2} \geq 100 \Longrightarrow x = 14$.
    - Step 4. Drop Egg 1 from floor 14. If it breaks, test floors 1–13 sequentially with Egg 2 (max 14 drops).

- Step 5. If it survives, test floor $14 + 13 = 27$, then floor $27 + 12 = 39$, and so on. The maximum required drops in the worst case is 14.

- **Animated Video:** Can you solve the egg drop riddle? - Yossi Elran

8. The Wildebeest & Lions River Crossing (State Space Search)

- **Problem Statement:** 3 lions and 3 wildebeest must cross a river using a boat that holds at most 2 animals at a time. At least 1 animal must row the boat back. If lions ever outnumber wildebeest on either bank, the lions will attack the wildebeest.

- **Question:** What sequence of boat trips safely moves all 6 animals across?

- **Answer & Explanation:**

  - Step 1. Trip 1: 2 lions cross; 1 lion returns (1 lion on far bank, 2 lions & 3 wildebeest on start bank).

  - Step 2. Trip 2: 2 lions cross; 1 lion returns (2 lions on far bank, 1 lion & 3 wildebeest on start bank).

  - Step 3. Trip 3: 2 wildebeest cross; 1 lion and 1 wildebeest return (1 lion & 1 wildebeest on far bank, 2 lions & 2 wildebeest on start bank).

  - Step 4. Trip 4: 2 wildebeest cross; 1 lion returns (3 wildebeest on far bank, 3 lions on start bank).

  - Step 5. Trip 5: 2 lions cross; 1 lion returns, then final 2 lions cross. All 6 cross safely in 11 trips.

- **Animated Video:** Can you solve the river crossing riddle? - Lisa Winer

9. The Coliseum Guard Passcode (Boolean Logic / Truth Tables)

- **Problem Statement:** Captured prisoners must select the correct escape door out of 3 doors based on clues given by 3 guards.

  - Guard A always tells the truth.

  - Guard B always lies.

  - Guard C answers randomly.

- You do not know which guard is which, and they only answer "Yes" or "No".

- **Question:** What structural logical question guarantees revealing a trustworthy guard to identify the correct door?

- **Answer & Explanation:**

  - Step 1. Ask a nested boolean question to neutralize the truth-teller vs. liar distinction: *"If I asked you whether Door 1 is safe, would you say 'Yes'?"*

  - Step 2. Truth-teller answers truthfully ("Yes"), while the Liar double-negates his lie and also answers "Yes".

  - Step 3. Direct the first question away from the random guard (Guard C) by identifying a non-random guard first.

  - Step 4. Evaluating compound logic eliminates ambiguity and isolates the safe door in 3 targeted questions.

- **Animated Video:** Can you solve the passcode riddle? - Ganesh Pai

10. The Space Monolith Probe (Finite State Automata)

- **Problem Statement:** A research probe must pass through 4 hazardous atmospheric layers surrounding an alien monolith. The probe can be coated with 1 of 3 protective shields: Red, Purple, or Green.

  - Layer 1 degrades Red coatings.

  - Layer 2 degrades Purple coatings.

  - Layer 3 degrades Green coatings.

  - Re-applying a coating takes time, and no two consecutive layers can use the same color.

- **Question:** How can you sequence the shield applications so the probe reaches the monolith undamaged?

- **Answer & Explanation:**

  - Step 1. Represent the rules as a state transition diagram where each layer restricts valid outgoing state transitions.

- Step 2. Layer 1 restricts Red ⟹ choose Green or Purple.
- Step 3. Layer 2 restricts Purple and forbids previous color ⟹ transition to Red.
- Step 4. Layer 3 restricts Green and forbids Red ⟹ transition to Purple.
- Step 5. Layer 4 permits Green ⟹ transition to Green. Sequence: Green -> Red -> Purple -> Green.

- **Animated Video:** Can you solve the alien probe riddle? - Dan Finkel

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